

Fundamentals of Electric Circuits

电路基础

Lecture10: Sinusoidal Steady-State Analysis

AC Power

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Outline

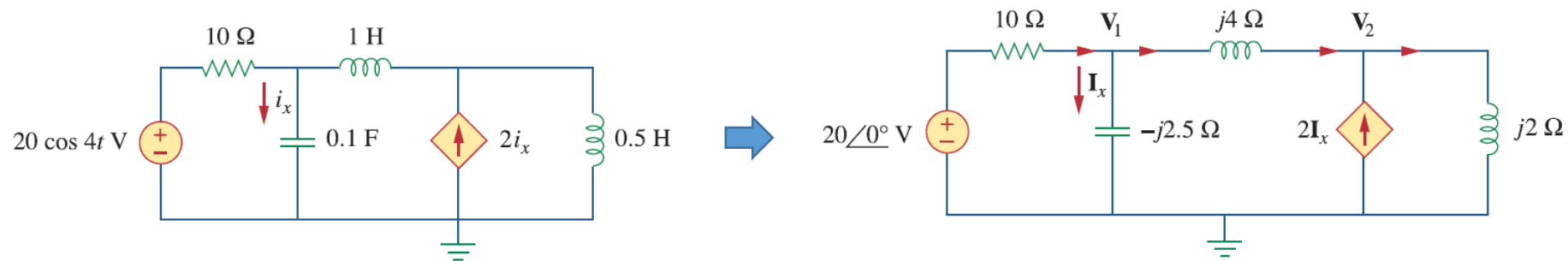
- **Sinusoidal Steady-State Analysis**
 - Nodal Analysis, Mesh Analysis
 - Superposition Theorem
 - Source Transformation
 - Thevenin and Norton Equivalent Circuits
- **AC Power Analysis**
 - Instantaneous and Average Power
 - Maximum Average Power Transfer
 - Effective Value and RMS Value
 - Apparent Power and Power Factor, Complex Power

Sinusoidal Steady-State Analysis

- There are three steps to analyzing an AC circuit
 - 1). Transform the circuit into the phasor or frequency domain
 - 2). Solve the problem using circuit techniques (in phasor form)
 - 3). Transform back to time domain
- Making use the generalized Ohm's law, the procedure will be simpler than direct analysis in time domain
- Techniques developed for DC circuits can equally applied

Sinusoidal Steady-State Analysis: Nodal Analysis (I)

- Example: Find $i_x(t)$ in the circuit using nodal analysis



Solution:

We first convert the circuit to the frequency domain:

$$\begin{aligned} 20 \cos 4t &\Rightarrow 20 \angle 0^\circ, & \omega &= 4 \text{ rad/s} \\ 1 \text{ H} &\Rightarrow j\omega L = j4 \\ 0.5 \text{ H} &\Rightarrow j\omega L = j2 \\ 0.1 \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j2.5 \end{aligned}$$

Sinusoidal Steady-State Analysis: Nodal Analysis (II)

- Example: Find $i_x(t)$ in the circuit using nodal analysis

Applying KCL at node 1,

$$\frac{20 - \mathbf{V}_1}{10} = \frac{\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4}$$

or

$$(1 + j1.5)\mathbf{V}_1 + j2.5\mathbf{V}_2 = 20$$

At node 2,

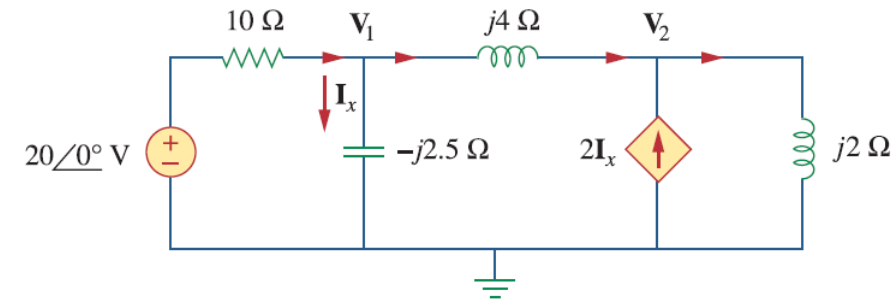
$$2\mathbf{I}_x + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

But $\mathbf{I}_x = \mathbf{V}_1 / -j2.5$. Substituting this gives

$$\frac{2\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

By simplifying, we get

$$11\mathbf{V}_1 + 15\mathbf{V}_2 = 0$$



$$\mathbf{V}_1 = 18.97 \angle 18.43^\circ \text{ V}$$

$$\mathbf{V}_2 = 13.91 \angle 198.3^\circ \text{ V}$$

$$\mathbf{I}_x = \frac{\mathbf{V}_1}{-j2.5} = 7.59 \angle 108.4^\circ \text{ A}$$

$$i_x = 7.59 \cos(4t + 108.4^\circ) \text{ A}$$

Sinusoidal Steady-State Analysis: Mesh Analysis

- Q: Determine current $\mathbf{I}_o$ in the circuit using mesh analysis

Solution:

Applying KVL to mesh 1, we obtain

$$(8 + j10 - j2)\mathbf{I}_1 - (-j2)\mathbf{I}_2 - j10\mathbf{I}_3 = 0$$

For mesh 2,

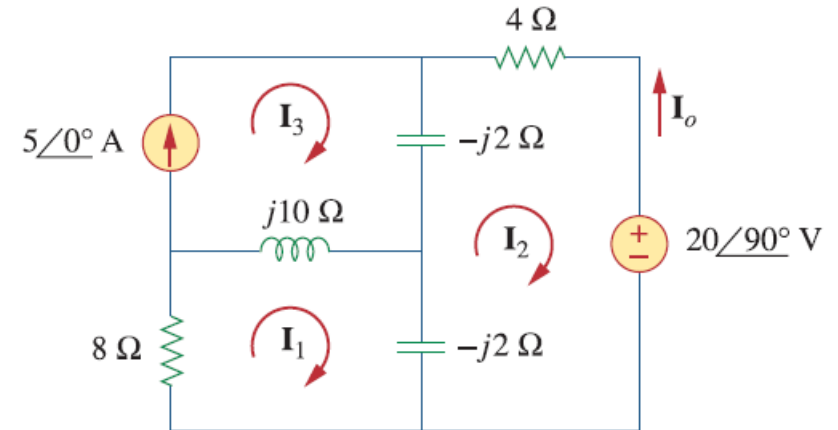
$$(4 - j2 - j2)\mathbf{I}_2 - (-j2)\mathbf{I}_1 - (-j2)\mathbf{I}_3 + 20\angle 90^\circ = 0$$

For mesh 3, $\mathbf{I}_3 = 5$.

$$\begin{aligned} \Rightarrow (8 + j8)\mathbf{I}_1 + j2\mathbf{I}_2 &= j50 \\ j2\mathbf{I}_1 + (4 - j4)\mathbf{I}_2 &= -j20 - j10 \end{aligned}$$

$$\Rightarrow \mathbf{I}_2 = 6.12\angle -35.22^\circ \text{ A}$$

$$\mathbf{I}_o = -\mathbf{I}_2 = 6.12\angle 144.78^\circ \text{ A}$$

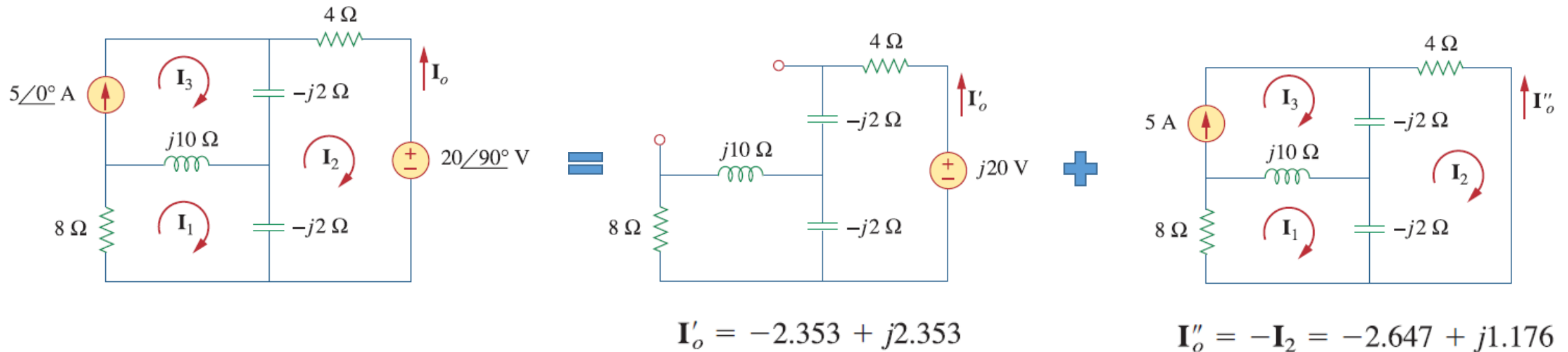


Superposition Theorem (I)

- Since AC circuits are **linear**, it is also possible to apply the principle of superposition
- Particular attention is needed when the circuit has various sources that **operate at different frequencies** → The complication is that, corresponding to each source, the circuit would have distinct equivalents in the frequency (phasor) domain
- The reason for this is that:
 - 1) the circuit elements may have a frequency-dependent impedances
 - 2) the resulting voltages/ currents corresponding to each frequencies must be converted back to the time domain before they can be added together
- This is because there is an **exponential factor $e^{j\omega t}$ implicit** in the sinusoidal analysis

Superposition Theorem (II)

- Use the superposition theorem to find I_o in the circuit.

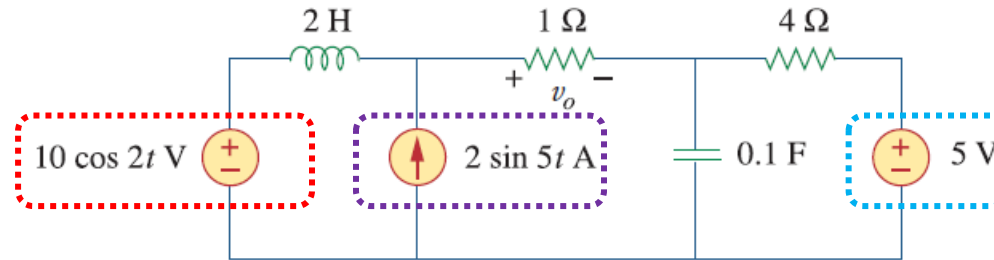


- Note that the circuit has sources operating at the same frequency → Can sum up in phasor domain

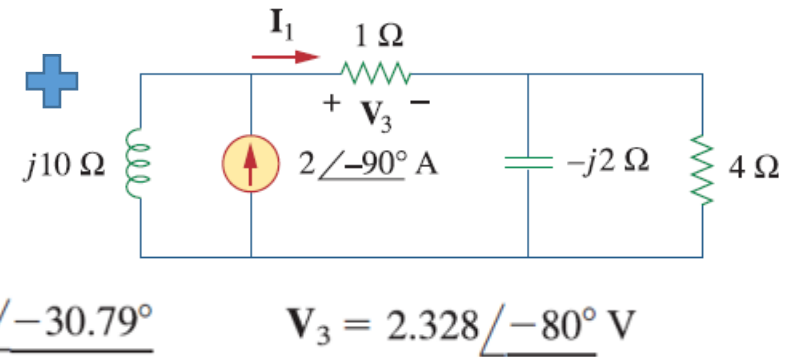
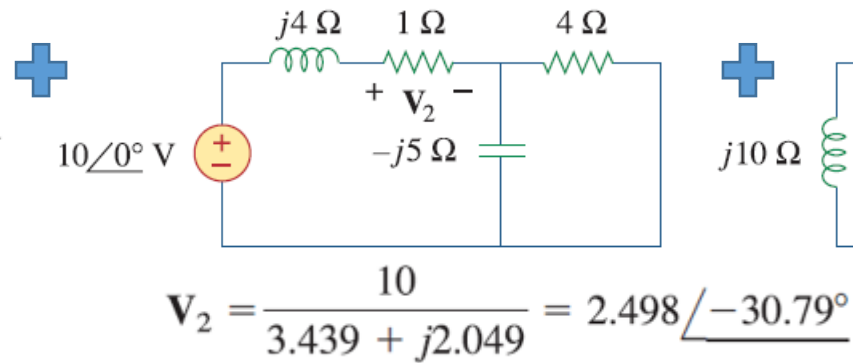
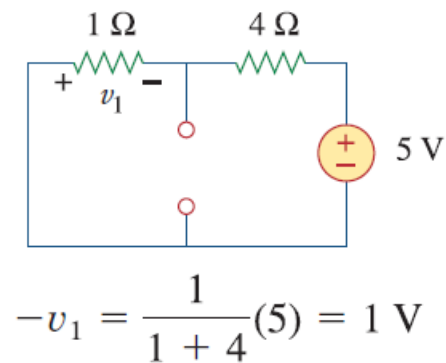
$$I_o = I'_o + I''_o = -5 + j3.529 = 6.12\angle 144.78^\circ\text{ A}$$

Superposition Theorem (III)

- Q: Find V_o of the circuit using the superposition theorem. Note that the circuit has sources operating at the different frequencies



- Solve in phasor domain



$$v_2 = 2.498 \cos(2t - 30.79^\circ)$$

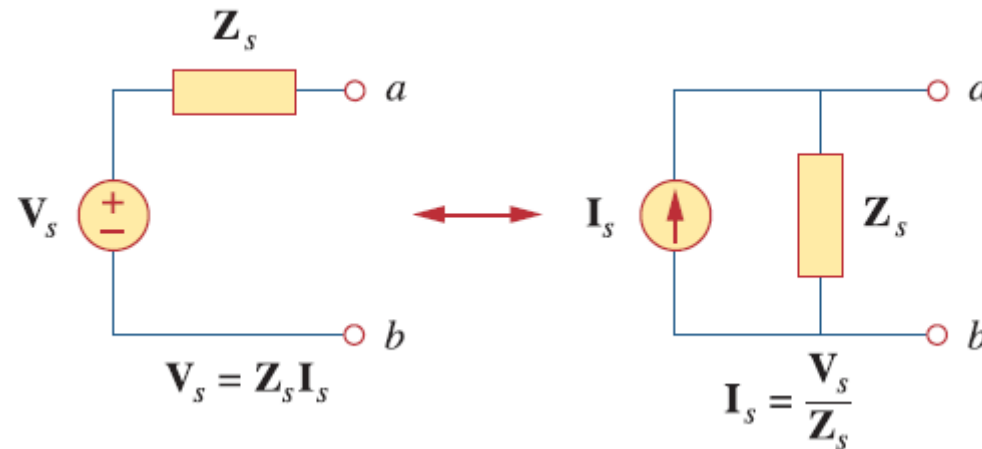
$$v_3 = 2.33 \cos(5t - 80^\circ) = 2.33 \sin(5t + 10^\circ) \text{ V}$$

- Adding up in time domain

$$v_o(t) = -1 + 2.498 \cos(2t - 30.79^\circ) + 2.33 \sin(5t + 10^\circ) \text{ V}$$

Source Transformation

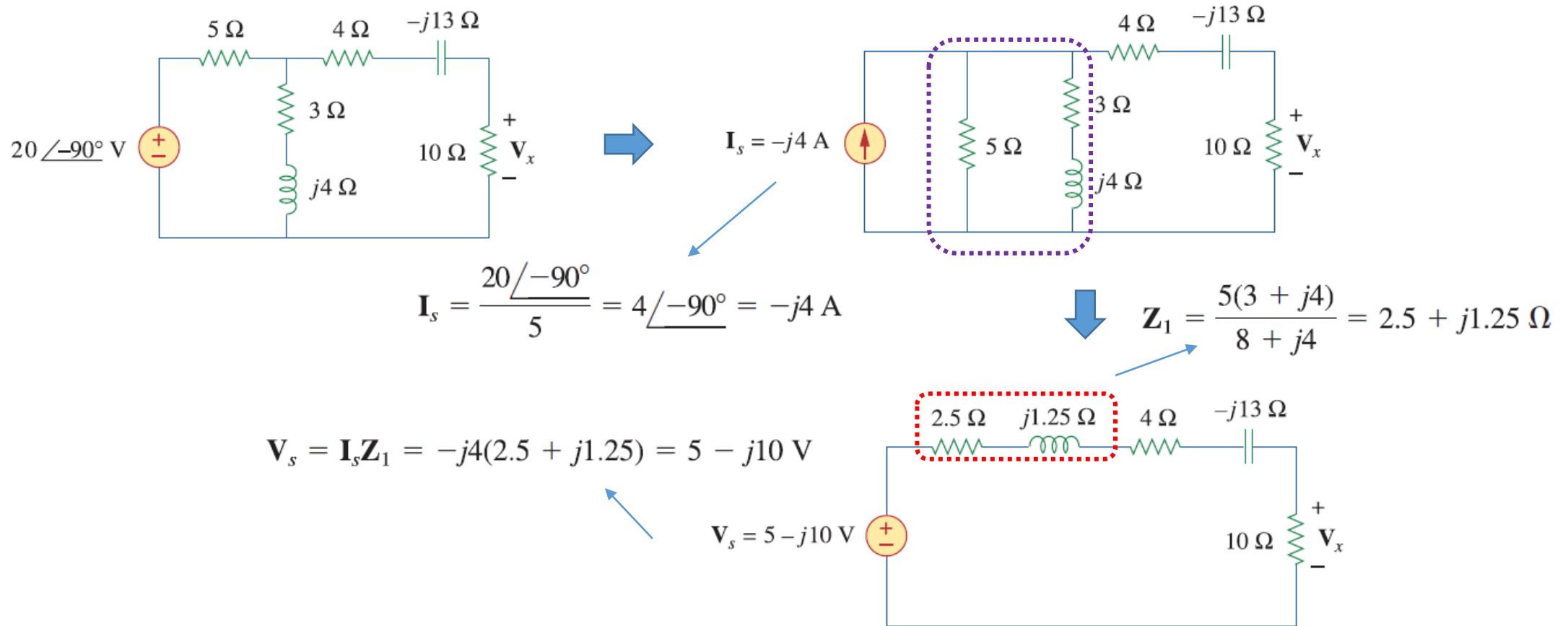
- Same as the case with DC sources, source transformation of circuits excited by AC sources in frequency domain involves transforming a voltage source in series with an impedance to a current source in parallel with an impedance



$$V_s = Z_s I_s \quad \Leftrightarrow \quad I_s = \frac{V_s}{Z_s}$$

Example: Source Transformation

- Q: Calculate V_x in the circuit using the concept of source transformation

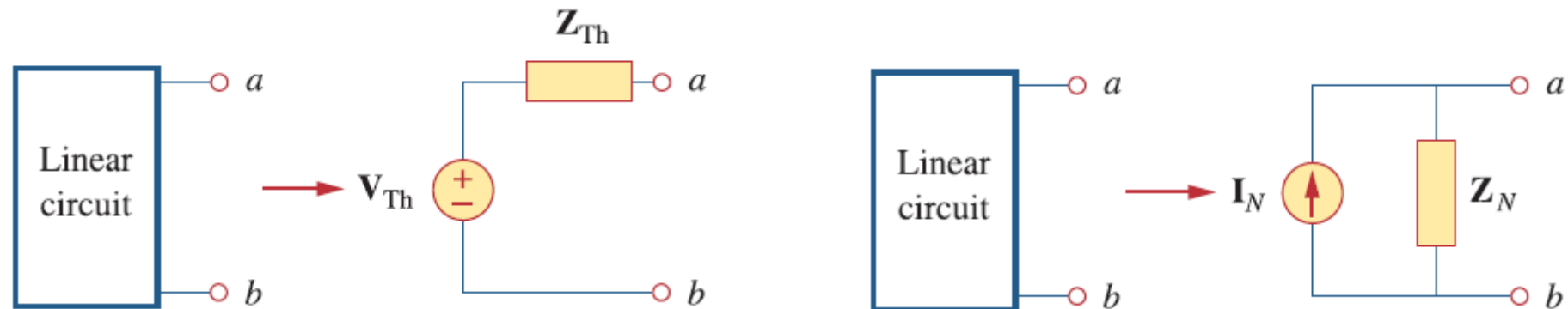


- By voltage division

$$V_x = \frac{10}{10 + 2.5 + j1.25 + 4 - j13} (5 - j10) = 5.519\angle -28^\circ\text{ V}$$

Thevenin and Norton Equivalent Circuits

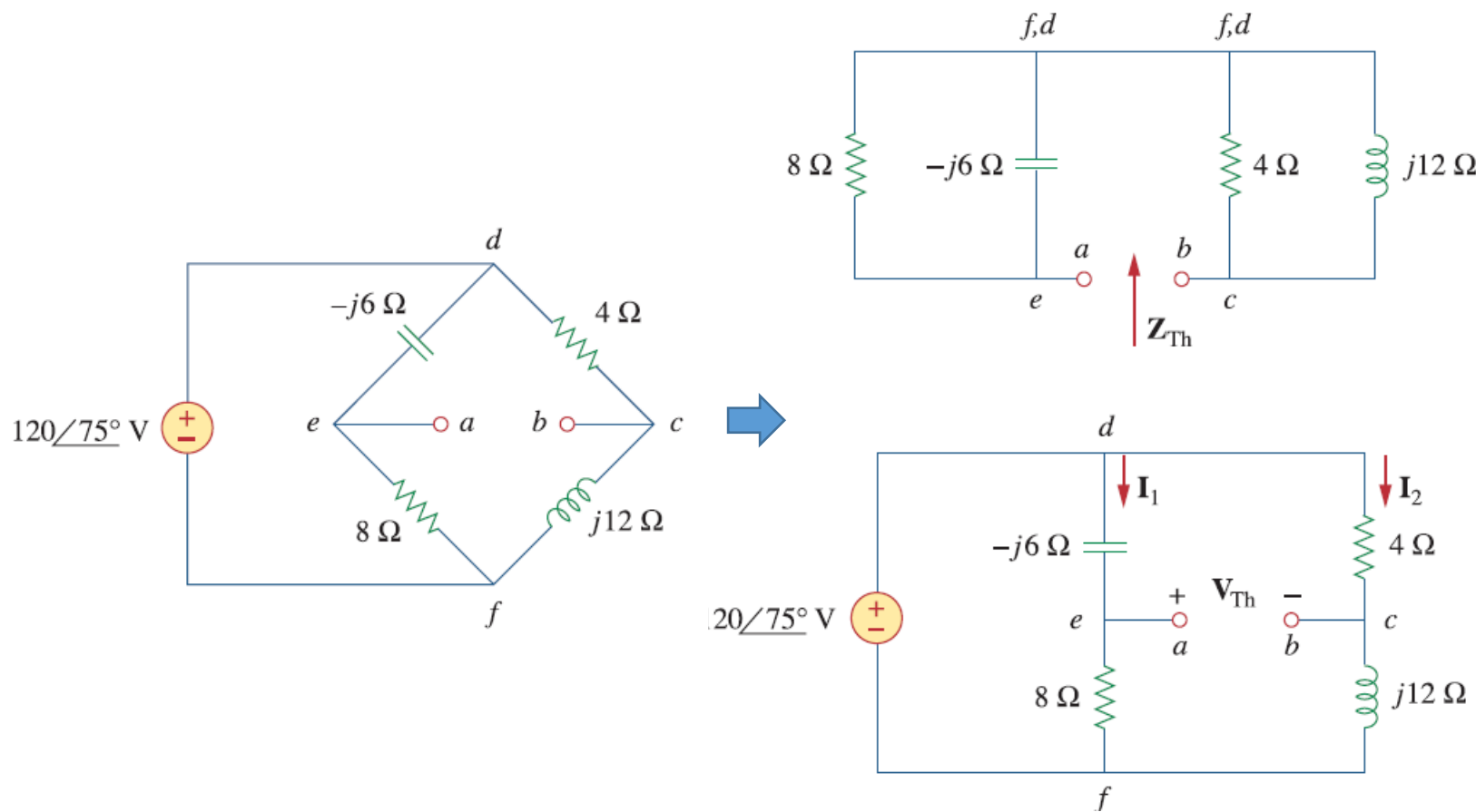
- Both Thevenin and Norton's theorems are applicable to AC circuits in phasor domain in the same way as DC. The only difference is the fact that the calculated values will be of complex number



$$V_{Th} = Z_N I_N, \quad Z_{Th} = Z_N$$

Example: Thevenin and Norton Equivalency

- Q: Obtain the Thevenin equivalent at terminals $a-b$



$$Z_1 = -j6 \parallel 8 = \frac{-j6 \times 8}{8 - j6} = 2.88 - j3.84\ \Omega$$

$$Z_2 = 4 \parallel j12 = \frac{j12 \times 4}{4 + j12} = 3.6 + j1.2\ \Omega$$

$$Z_{Th} = Z_1 + Z_2 = 6.48 - j2.64\ \Omega$$

$$I_1 = \frac{120\angle 75^\circ}{8 - j6} \text{ A}, \quad I_2 = \frac{120\angle 75^\circ}{4 + j12} \text{ A}$$

$$V_{Th} - 4I_2 + (-j6)I_1 = 0$$

$$\begin{aligned} V_{Th} &= -28.936 - j24.55 \\ &= 37.95\angle 220.31^\circ \text{ V} \end{aligned}$$

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AC Power Analysis

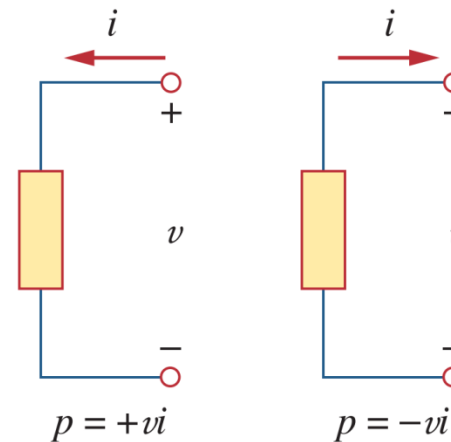
- Review:** Power is defined as the rate of energy provided or consumed by a device per unit time. It is measured in Watt (W)

$$p \triangleq \frac{dw}{dt}$$

$$w = \int_{t_0}^t p \, dt = \int_{t_0}^t vi \, dt$$

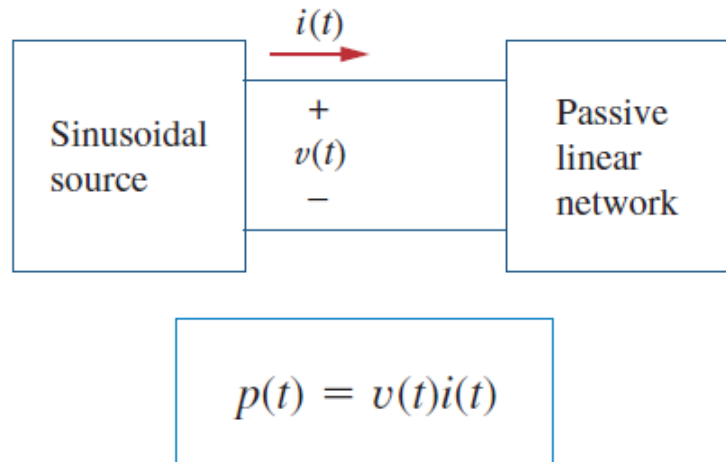
- It is equal to the product of voltage and current

$$p = vi$$



Instantaneous Power (I)

- The **instantaneous power** (瞬时功率) is the power at any instant of time



$$v(t) = V_m \cos(\omega t + \theta_v)$$

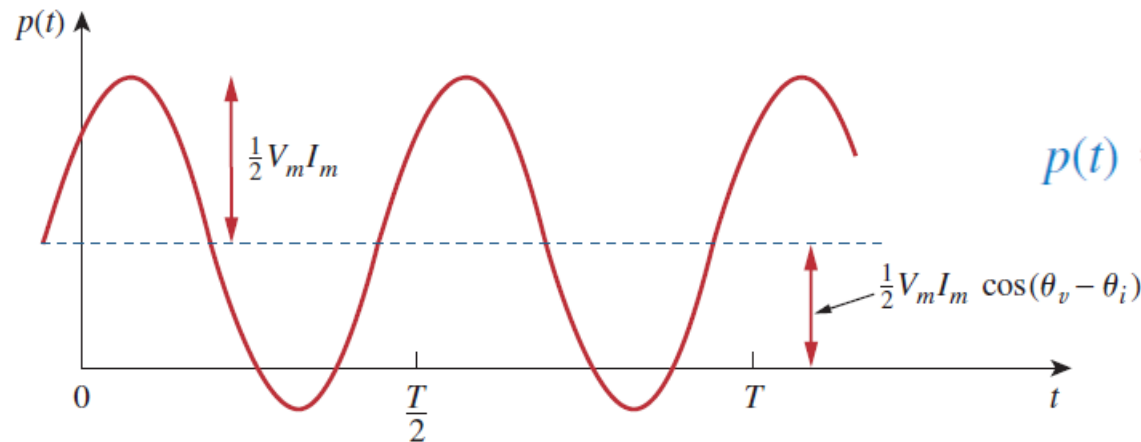
$$i(t) = I_m \cos(\omega t + \theta_i)$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$



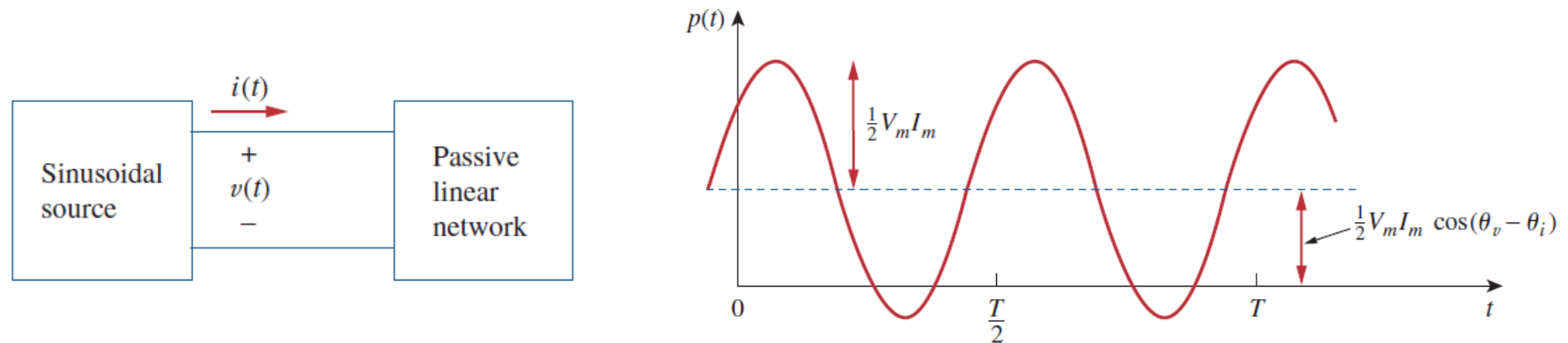
$$p(t) = v(t)i(t) = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$



Instantaneous Power (II)

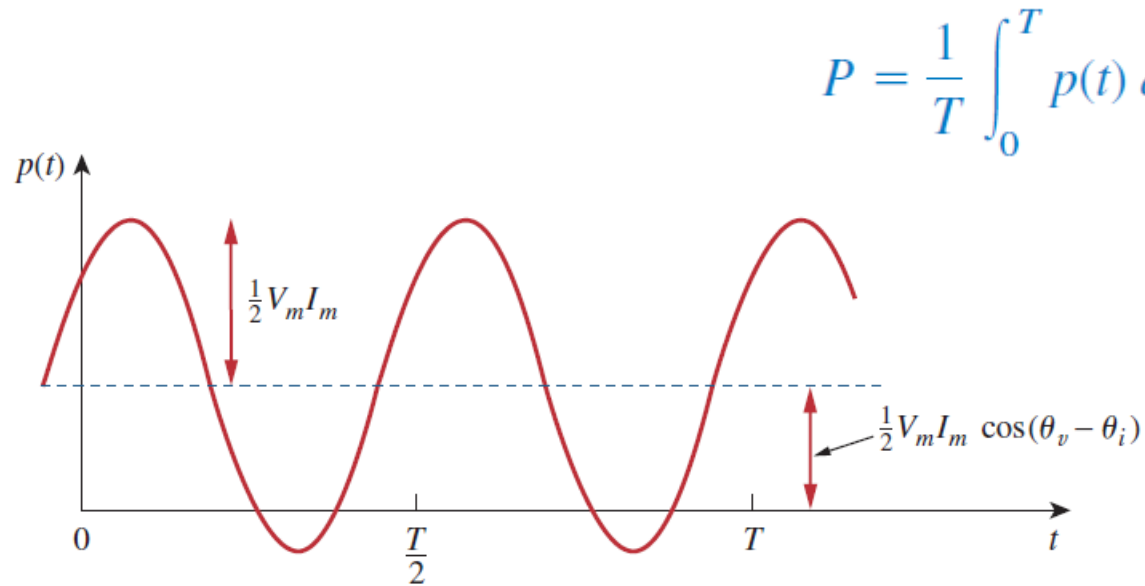
$$p(t) = \boxed{\frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)} + \boxed{\frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)}$$



- Note that the power is consist of two components
 - A constant dependent on the phase difference between the voltage and current
 - A sinusoid oscillating at twice the frequency of the voltage/current

Average Power (I)

- The **average power** (平均功率) is the average of the instantaneous power over one period. The average power is given by



$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

$$P = \frac{1}{T} \int_0^T p(t) dt \quad \Rightarrow \quad P = \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) dt + \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i) dt$$

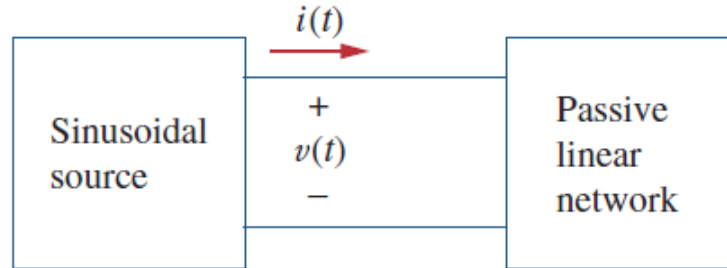
$$= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \frac{1}{T} \int_0^T dt + \frac{1}{2} V_m I_m \frac{1}{T} \int_0^T \cos(2\omega t + \theta_v + \theta_i) dt$$

~~$$+ \frac{1}{2} V_m I_m \frac{1}{T} \int_0^T \cos(2\omega t + \theta_v + \theta_i) dt$$~~



$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

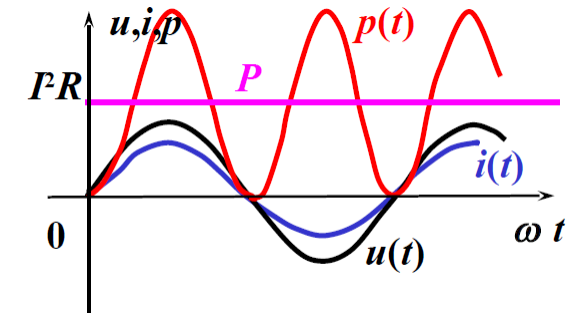
Average Power (II)



$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

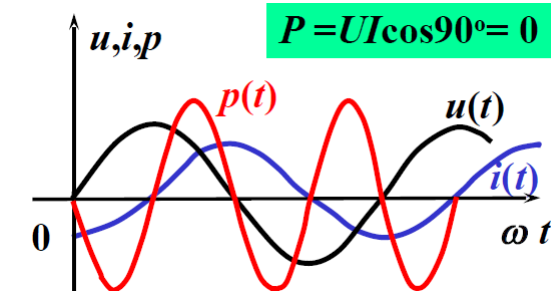
- When $\theta_v = \theta_i$, the voltage and current are in phase

$$P = \frac{1}{2} V_m I_m = \frac{1}{2} I_m^2 R = \frac{1}{2} |\mathbf{I}|^2 R$$



- When $\theta_v - \theta_i = \pm 90^\circ$, we have a purely reactive circuit, and

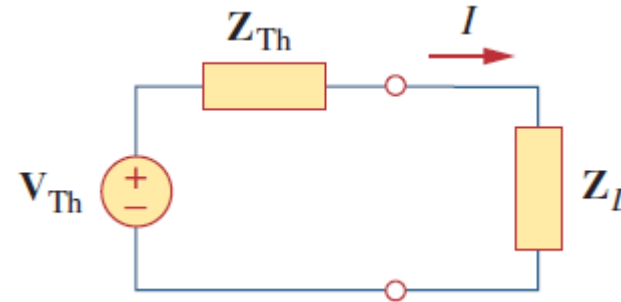
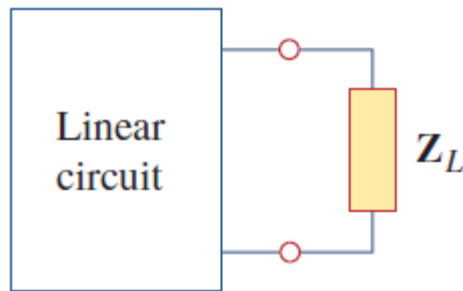
$$P = \frac{1}{2} V_m I_m \cos 90^\circ = 0$$



A resistive load (R) absorbs power at all times, while a reactive load (L or C) absorbs zero average power.

Maximum Average Power Transfer (I)

- For **maximum average power transfer** (最大平均功率传输), the load impedance Z_L must be equal to the **complex conjugate** (复共轭) of the Thevenin impedance Z_{Th}



$$Z_{Th} = R_{Th} + jX_{Th}$$

$$Z_L = R_L + jX_L$$

$$I = \frac{V_{Th}}{Z_{Th} + Z_L} = \frac{V_{Th}}{(R_{Th} + jX_{Th}) + (R_L + jX_L)}$$

$$P = \frac{1}{2}|I|^2 R_L = \frac{|V_{Th}|^2 R_L / 2}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}$$

To find the
maximum



$$\frac{\partial P}{\partial X_L} = -\frac{|V_{Th}|^2 R_L (X_{Th} + X_L)}{[(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2]^2}$$

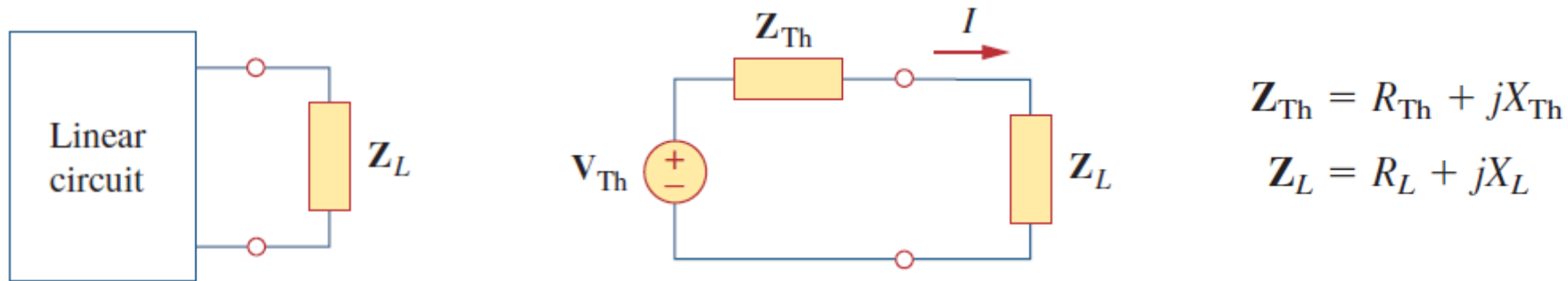
$$\frac{\partial P}{\partial R_L} = \frac{|V_{Th}|^2 [(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2 - 2R_L(R_{Th} + R_L)]}{2[(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2]^2}$$

Set to Zero

$$X_L = -X_{Th} \quad R_L = \sqrt{R_{Th}^2 + (X_{Th} + X_L)^2}$$

Maximum Average Power Transfer (II)

- For **maximum average power transfer** (最大平均功率传输), the load impedance Z_L must be equal to the **complex conjugate** (复共轭) of the Thevenin impedance Z_{Th}



$$Z_L = R_L + jX_L = R_{Th} - jX_{Th} = Z_{Th}^*$$

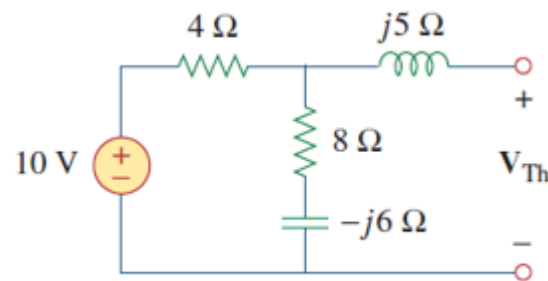
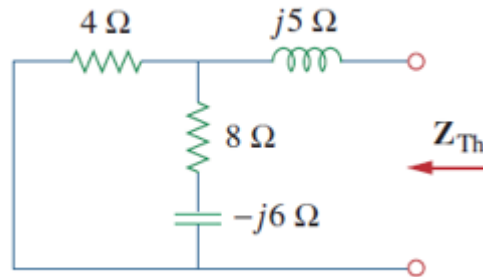
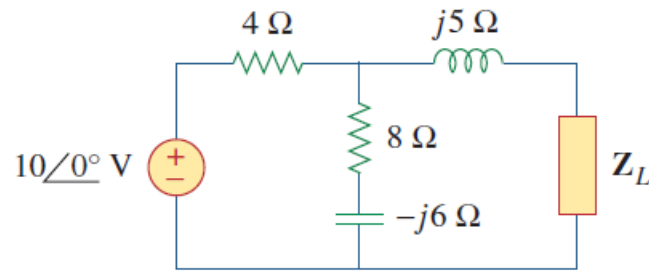
$$P = \frac{1}{2} |\mathbf{I}|^2 R_L = \frac{|\mathbf{V}_{Th}|^2 R_L / 2}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}$$



$$P_{\max} = \frac{|\mathbf{V}_{Th}|^2}{8R_{Th}}$$

Example: Maximum Average Power Transfer

- Q: Determine the load impedance $\mathbf{Z}_L$ that maximizes the average power drawn from the circuit. What is the maximum average power?



Solution:

First we obtain the Thevenin equivalent at the load terminals. To get $\mathbf{Z}_{Th}$, consider the circuit shown in Fig. 11.9(a). We find

$$\mathbf{Z}_{Th} = j5 + 4 \parallel (8 - j6) = j5 + \frac{4(8 - j6)}{4 + 8 - j6} = 2.933 + j4.467 \, \Omega$$

$$\mathbf{V}_{Th} = \frac{8 - j6}{4 + 8 - j6}(10) = 7.454 \angle -10.3^\circ \, \text{V}$$

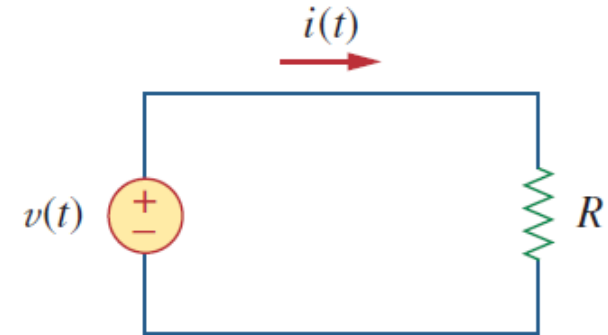
$$\mathbf{Z}_L = \mathbf{Z}_{Th}^* = 2.933 - j4.467 \, \Omega$$

$$P_{\max} = \frac{|\mathbf{V}_{Th}|^2}{8R_{Th}} = \frac{(7.454)^2}{8(2.933)} = 2.368 \, \text{W}$$

Effective Value and RMS Value (I)

- The **effective value** (有效值) of a period current is the dc current that delivers the same average power to a resistor as the period current
- Average power of periodic source

$$P = \frac{1}{T} \int_0^T i^2 R dt = \frac{R}{T} \int_0^T i^2 dt$$

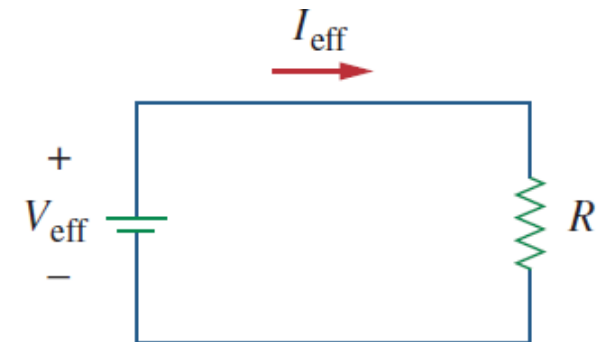


- Equivalent DC current delivering the same power

$$P = I_{\text{eff}}^2 R$$

- Equivalent current/voltage

$$I_{\text{eff}} = \sqrt{\frac{1}{T} \int_0^T i^2 dt} \quad V_{\text{eff}} = \sqrt{\frac{1}{T} \int_0^T v^2 dt}$$



- The effective value of a periodic signal is its **root mean square (rms, 方均根值)** value

$$X_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T x^2 dt}$$

Effective Value and RMS Value (II)

- For the sinusoid $i(t) = I_m \cos \omega t$

$$\begin{aligned} I_{\text{rms}} &= \sqrt{\frac{1}{T} \int_0^T I_m^2 \cos^2 \omega t \, dt} \\ &= \sqrt{\frac{I_m^2}{T} \int_0^T \frac{1}{2} (1 + \cos 2\omega t) \, dt} = \frac{I_m}{\sqrt{2}} \end{aligned}$$

- Similarly, for $v(t) = V_m \cos \omega t$

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$$

- Now the average power can be written in terms of rms value

$$\begin{aligned} P &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) \quad \text{Or} \quad P = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R} \\ &= V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) \end{aligned}$$

Example: Effective and RMS Value

Determine the rms value of the current waveform in Fig. 11.14. If the current is passed through a $2\text{-}\Omega$ resistor, find the average power absorbed by the resistor.

Solution:

The period of the waveform is $T = 4$. Over a period, we can write the current waveform as

$$i(t) = \begin{cases} 5t, & 0 < t < 2 \\ -10, & 2 < t < 4 \end{cases}$$

The rms value is

$$\begin{aligned} I_{\text{rms}} &= \sqrt{\frac{1}{T} \int_0^T i^2 dt} = \sqrt{\frac{1}{4} \left[\int_0^2 (5t)^2 dt + \int_2^4 (-10)^2 dt \right]} \\ &= \sqrt{\frac{1}{4} \left[25 \frac{t^3}{3} \Big|_0^2 + 100t \Big|_2^4 \right]} = \sqrt{\frac{1}{4} \left(\frac{200}{3} + 200 \right)} = 8.165 \text{ A} \end{aligned}$$

The power absorbed by a $2\text{-}\Omega$ resistor is

$$P = I_{\text{rms}}^2 R = (8.165)^2 (2) = 133.3 \text{ W}$$

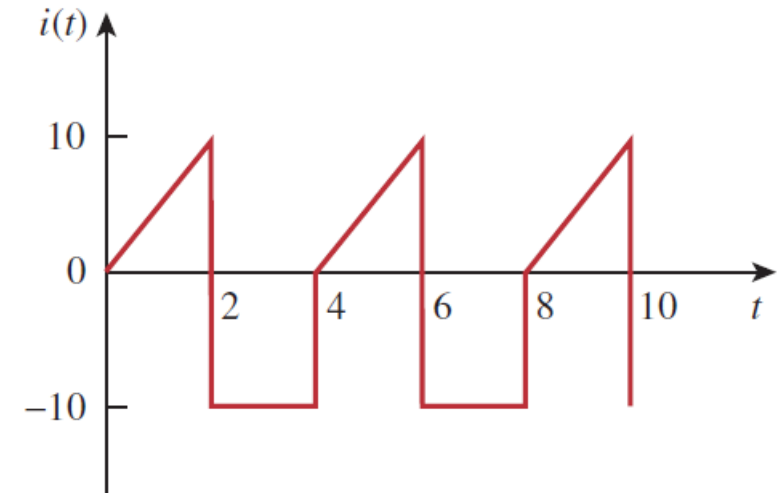


Figure 11.14

- Look again the formula for average power. It is a sinusoid with amplitude specified by $V_{rms}I_{rms}$, and argument specified by $\theta_v - \theta_i$

$$\begin{aligned} P &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) \\ &= V_{rms} I_{rms} \cos(\theta_v - \theta_i) \end{aligned}$$

- The **apparent power** (视在功率), denote it as **S**, is the product of the rms values of voltage and current

$$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i) = S \cos(\theta_v - \theta_i)$$

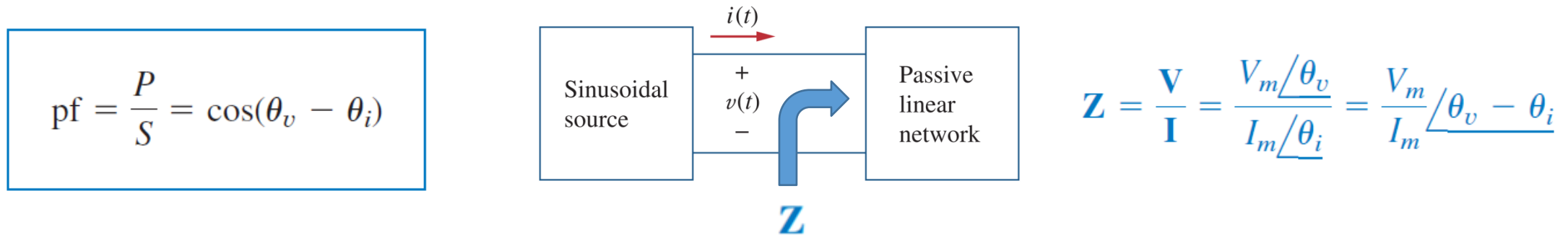
$$S = V_{rms} I_{rms}$$

- Also, the ratio between the average power to the apparent power as **power factor** (pf, 功率因数)

$$\text{pf} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

Apparent Power and Power Factor (II)

- The ratio between the average power to the apparent power as power factor (pf)
- The **power factor** is the **cosine of the phase difference between voltage and current**. It is also the **cosine of the angle of the load impedance**



- The power factor takes value in the range of $[0, 1]$
- pf = 1** for purely resistive loads
- pf = 0** for purely reactive (inductive or capacitive) loads
- Power factors affect the electricity bills – low power factor needs to pay more

Example: Apparent Power and Power Factor

- Q: A series-connected load draws a current $i(t) = 4 \cos(100\pi t + 10^\circ)$ A when the applied voltage is $v(t) = 120 \cos(100\pi t - 20^\circ)$ V. Find the apparent power and the power factor of the load. Determine the element values that form the series-connected load.

Solution:

The apparent power is

$$S = V_{\text{rms}} I_{\text{rms}} = \frac{120}{\sqrt{2}} \frac{4}{\sqrt{2}} = 240 \text{ VA}$$

The power factor is

$$\text{pf} = \cos(\theta_v - \theta_i) = \cos(-20^\circ - 10^\circ) = 0.866 \quad (\text{leading})$$

The pf is leading because the current leads the voltage. The pf may also be obtained from the load impedance.

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{120 \angle -20^\circ}{4 \angle 10^\circ} = 30 \angle -30^\circ = 25.98 - j15 \, \Omega$$

$$\text{pf} = \cos(-30^\circ) = 0.866 \quad (\text{leading})$$

The load impedance $\mathbf{Z}$ can be modeled by a 25.98- Ω resistor in series with a capacitor with

$$X_C = -15 = -\frac{1}{\omega C}$$

or

$$C = \frac{1}{15\omega} = \frac{1}{15 \times 100\pi} = 212.2 \, \mu\text{F}$$

Complex Power (I)

- So far, we carry out our discussion mainly in time domain. We have learned that representing voltage and current in frequency (phasor) domain usually leads to a much simpler treatment. Now we do the same for power

- In time domain

$$v(t) = V_m \cos(\omega t + \theta_v) \quad \text{and} \quad i(t) = I_m \cos(\omega t + \theta_i)$$

$$P = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

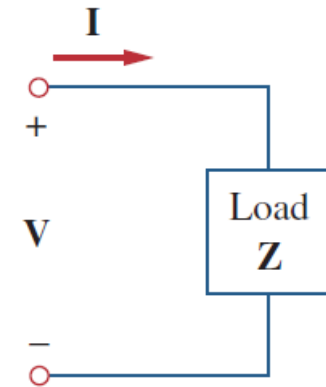
- In phasor domain

$$\mathbf{V} = V_m \angle \theta_v \quad \text{and} \quad \mathbf{I} = I_m \angle \theta_i$$

- In analogy to voltage and current, we define a new phasor for power: **complex power** (复功率)

$$\mathbf{S} = \frac{1}{2} \mathbf{V} \mathbf{I}^* = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$

➤ Where $\mathbf{V}_{\text{rms}} = \frac{\mathbf{V}}{\sqrt{2}} = V_{\text{rms}} \angle \theta_v$ $\mathbf{I}_{\text{rms}} = \frac{\mathbf{I}}{\sqrt{2}} = I_{\text{rms}} \angle \theta_i$ are effective voltage and current phasors



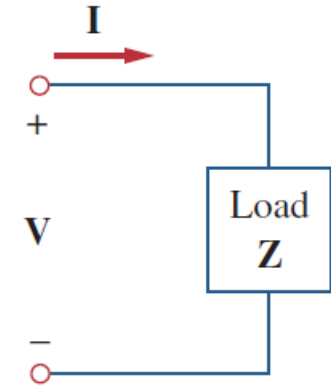
Complex Power (II)

- We can rewrite the complex power as

$$\begin{aligned} \mathbf{S} &= V_{\text{rms}} I_{\text{rms}} \angle \theta_v - \theta_i \\ &= V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) + j V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i) \end{aligned}$$



$$\mathbf{S} = I_{\text{rms}}^2 \mathbf{Z} = \frac{V_{\text{rms}}^2}{\mathbf{Z}^*} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$



$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{V_m \angle \theta_v}{I_m \angle \theta_i} = \frac{V_m}{I_m} \angle \theta_v - \theta_i$$

- Since $\mathbf{Z} = R + jX$, then

$$\mathbf{S} = I_{\text{rms}}^2 (R + jX) = P + jQ$$

$$P = \text{Re}(\mathbf{S}) = I_{\text{rms}}^2 R = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

$$Q = \text{Im}(\mathbf{S}) = I_{\text{rms}}^2 X = V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i)$$

- Where P is the **average or real power** (有用功率), and Q is called **reactive (or quadrature) power** (无功功率), related to the load's reactance

Real and Reactive Power

- P is the average or real power (有用功率), and Q is called reactive (or quadrature) power (无功功率), related to the load's reactance
- The real power P is the only useful power. It is measured in watts
- The reactive power Q is a measure of the energy exchange between the source and the reactive load. It is measured in units of volt-ampere reactive (VAR, 无功伏安)

1. $Q = 0$ for resistive loads (unity pf).
2. $Q < 0$ for capacitive loads (leading pf).
3. $Q > 0$ for inductive loads (lagging pf).

Complex power (in VA) is the product of the rms voltage phasor and the complex conjugate of the rms current phasor. As a complex quantity, its real part is real power P and its imaginary part is reactive power Q .

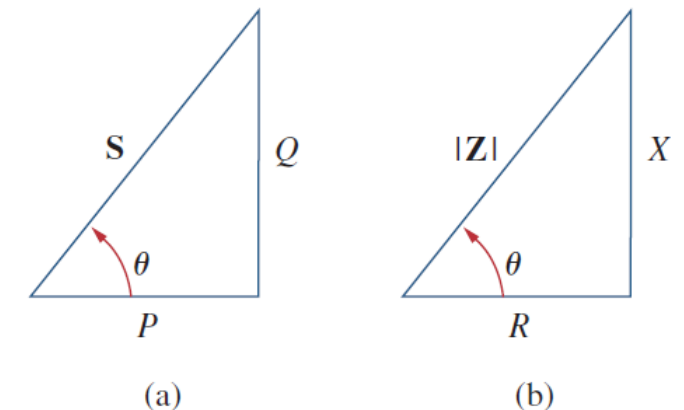


Figure 11.21
(a) Power triangle, (b) impedance triangle.

Example: Real and Reactive Power

- Q: The voltage across a load is $v(t) = 60 \cos(\omega t - 10^\circ)$ V and the current through the element in the direction of the voltage drop is $i(t) = 1.5 \cos(\omega t + 50^\circ)$ A. Find: a) the complex and apparent powers, b) the real and reactive powers and c) the power factor and the load impedance

Solution:

(a) For the rms values of the voltage and current, we write

$$\mathbf{V}_{\text{rms}} = \frac{60}{\sqrt{2}} \angle -10^\circ, \quad \mathbf{I}_{\text{rms}} = \frac{1.5}{\sqrt{2}} \angle +50^\circ$$

The complex power is

$$\mathbf{S} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^* = \left(\frac{60}{\sqrt{2}} \angle -10^\circ \right) \left(\frac{1.5}{\sqrt{2}} \angle -50^\circ \right) = 45 \angle -60^\circ \text{ VA}$$

The apparent power is

$$S = |\mathbf{S}| = 45 \text{ VA}$$

(b) We can express the complex power in rectangular form as

$$\mathbf{S} = 45 \angle -60^\circ = 45 [\cos(-60^\circ) + j \sin(-60^\circ)] = 22.5 - j38.97$$

Since $\mathbf{S} = P + jQ$, the real power is

$$P = 22.5 \text{ W}$$

while the reactive power is

$$Q = -38.97 \text{ VAR}$$

(c) The power factor is

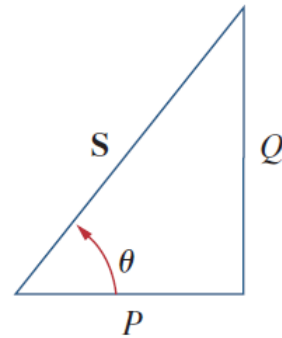
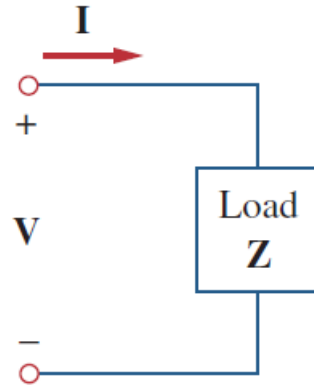
$$\text{pf} = \cos(-60^\circ) = 0.5 \text{ (leading)}$$

It is leading, because the reactive power is negative. The load impedance is

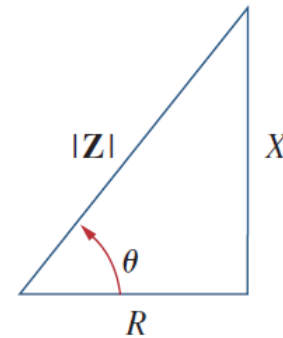
$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{60 \angle -10^\circ}{1.5 \angle +50^\circ} = 40 \angle -60^\circ \Omega$$

which is a capacitive impedance.

Summarizing AC Power



(a)



(b)

Figure 11.21

(a) Power triangle, (b) impedance triangle.

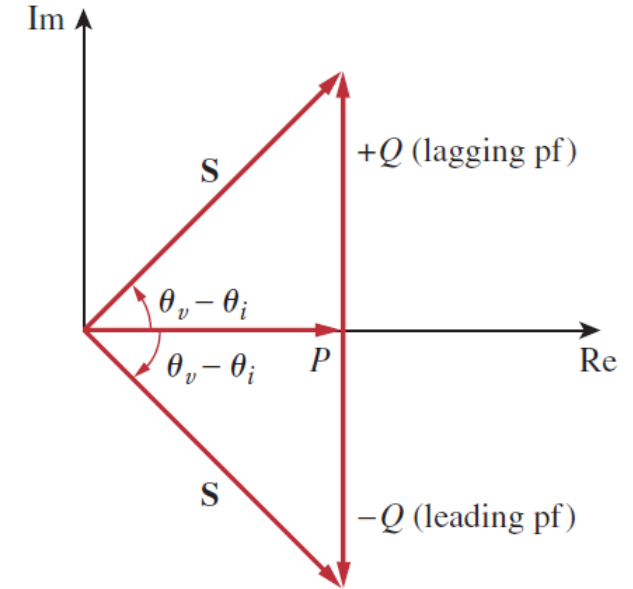


Figure 11.22

Power triangle.

$$\text{Complex Power} = \mathbf{S} = P + jQ = \mathbf{V}_{\text{rms}}(\mathbf{I}_{\text{rms}})^*$$

$$= |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| \angle \theta_v - \theta_i$$

$$\text{Apparent Power} = S = |\mathbf{S}| = |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| = \sqrt{P^2 + Q^2}$$

$$\text{Real Power} = P = \text{Re}(\mathbf{S}) = S \cos(\theta_v - \theta_i)$$

$$\text{Reactive Power} = Q = \text{Im}(\mathbf{S}) = S \sin(\theta_v - \theta_i)$$

$$\text{Power Factor} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

Summary

- The circuit law for DC can be equally applied for steady–state analysis of AC circuit
- Power can be analysis in either time domain or phasor domain. Phasor domain might be easy to calculate
- Power of AC circuit has two components: average DC term and a sinusoid with frequency 2ω
- Effective value are defined for AC variables as its DC equivalent in terms of having the same consuming power. For sinusoidal signal, it is $1/\sqrt{2}$ of its peak values (in both time or phasor domains)
- The phasor representation of power is the complex power, whose real power is average power, imaginary power is reactive power, modulus is apparent power, argument is power factor angle, and cosine of the angle is called power factor